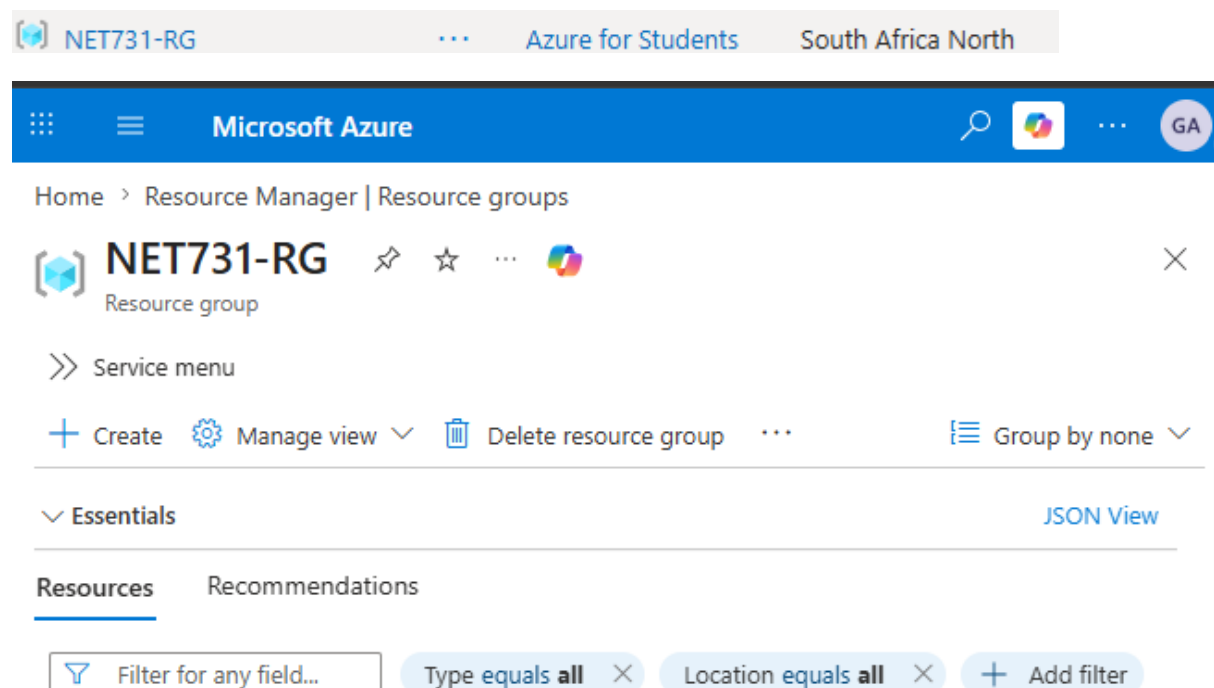


Week 2

Project overview

This week I worked on setting up the cloud infrastructure in Microsoft Azure. I made a Resource Group to keep all my resources in one place. I set up a Virtual Network and a subnet so everything can talk to each other. I also made a Network Security Group to control what comes in and out. Then I set up a Machine and connected it to all the network stuff I made. The whole point of this exercise was to make a safe and organized cloud environment that can handle all the things we will be doing on in the project, with Microsoft Azure. I used Microsoft Azure to build this cloud environment.

TASK 1 & 2



Create virtual network ...



Basics

Security

Address space

Tags

Review + create

Subscription *

Azure for Students



Resource group *

NET731-RG



[Create new](#)

Instance details

Virtual network name *

NET731-VNET

Region * ⓘ

(Africa) South Africa North

[Deploy to an Azure Extended Zone](#)

Name * ⓘ

subnet1

IPv4

Include an IPv4 address space



IPv4 address range ⓘ

10.0.0.0/16



10.0.0.0 - 10.0.255.255

Starting address * ⓘ

10.0.0.0



Size ⓘ

/24 (256 addresses)



Subnet address range ⓘ

10.0.0.0 - 10.0.0.255

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

20240435@ctucareer.co...
CTU CAREER (CTUCAREER.CO.ZA)

Home > Network foundation | Virtual networks

Create virtual network

Validation passed

BasicsSecurityAddress spaceTagsReview + create

Security

Azure Bastion	Disabled
Azure Firewall	Disabled
Azure DDoS Network Protection	Disabled

Address space

Address space	10.0.0.0/16 (65 536 addresses)
Subnet	subnet1 (10.0.0.0/24) (256 addresses)
- Private subnet	Enabled

Basics

Subscription	Azure for Students
Resource group	NET731-RG
Region	South Africa North
name	NET731-NSG

Microsoft Azure

GA

Home

CreateNetworkSecurityGroupBladeV2-20260...
Deployment

>> Service menu

Delete Cancel Redeploy Download Refresh

✓

Your deployment is complete

Deployment name : CreateNetworkSecurityGroupBladeV2-20260515131209

Subscription : Azure for Students

Resource group : NET731-RG

Start time : 5/15/2026, 1:14:37 PM

Correlation ID : b6fea48e-13f0-4e6b-8d52-8cc54b76de41

Task 3

Create network security group ...



Basics Tags Review + create

Project details

Subscription *	Azure for Students	▼
Resource group *	CTU-20240435-RG	▼

[Create new](#)

Instance details

Name *	TechBridge-NSG	✓
Region *	South Africa North	▼

Create network security group ...

✓ Validation passed

Basics Tags Review + create

Basics

Subscription	Azure for Students
Resource group	CTU-20240435-RG
Region	South Africa North
name	SecureTechBridge-NSG

SecureTechBridge-NSG
Network security group

How do I create an alert to track firewall metric failures

Search

Move Delete Refresh Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Inbound security rules

Outbound security rules

Network interfaces

Subnets

Properties

Locks

Monitoring

Essentials

Resource group (move) : [CTU-20240435-RG](#)

Location : South Africa North

Subscription (move) : [Azure for Students](#)

Subscription ID : 3eb7711b-c785-4870-b59d-5091741c08eb

Tags (edit) : [Add tags](#)

Custom security rules : 4 inbound, 0 outbound

Associated with : 0 subnets, 0 network interfaces

JSON View

Filter by name

Port == all Protocol == all Source == all Destination == all Action == all

Priority	Name	Port	Protocol	Source	Destination	Action
Inbound Security Rules						
100	Allow-HTTP	80	TCP	Any	Any	Allow
110	Allow-HTTPS	443	TCP	Any	Any	Allow
120	Allow-SSH-MyIP	22	TCP	41.56.157.224	Any	Allow
1000	Deny-All-Inbound	8080	Any	Any	Any	Deny
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow